

For a set $C \subset V$, the **polar cone** of C is

$$C^\circ = \{ y \in V^* : y(x) \leq 0 \quad \forall x \in C \} \subset V^*$$

From this definition, C° is a closed convex cone.

Theorem: For any C , $C^{\circ\circ} = \text{cl cone } C$.

Proof: First we show $C \subset C^{\circ\circ}$.

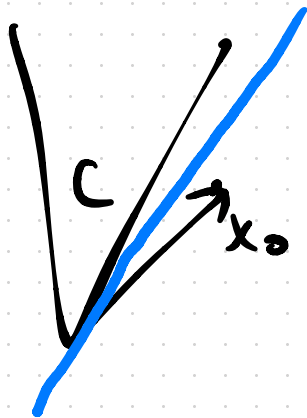
Suppose $x_0 \in C$. For any $y \in C^\circ$, $y(x_0) \leq 0$.

Thus $x_0 \in (C^\circ)^\circ$. So $C \subset C^{\circ\circ}$.

Since $C^{\circ\circ}$ is closed, $\text{cl } C \subset C^{\circ\circ}$.

Next show $C^{\circ\circ} \subset \text{cl } C$.

Suppose $x_0 \notin \text{cl } C$. By Farkas' lemma, $\exists y \in V^*$ such that $y(x_0) > 0$ and $y(x) \leq 0$ for all $x \in C$.



$y \in C^\circ$. Then since $y(x_0) > 0$,
 $x_0 \notin C^{\circ\circ}$. Therefore $C^{\circ\circ} \subset \text{cl } C$.

Theorem: If

(a)

$$C = \text{cone } S$$

then

$$C^\circ = \{ y \in V^* : y(x) \leq 0 \text{ for all } x \in S \} = S^\circ$$

(b) If

$$C = \{ x \in V : y_i(x) \leq 0 \text{ for all } i \in I \}$$

then

$$C^\circ = \text{cl cone } \{ y_i : i \in I \}$$

Proof:

(a) If $C = \text{cone } S$. $C^\circ = (\text{cone } S)^\circ \subset S^\circ$

If $y \in S^\circ$, then $y(x) \leq 0$ for all $x \in S$.

By linearity, $y(x) \leq 0$ for all $x \in \text{cone } S$.

So $y \in C^\circ$, so $C^\circ = S^\circ$.

(b) $C = (\{y_i\}_{i \in I})^\circ \Rightarrow C^\circ = (\{y_i\}_{i \in I})^{\circ\circ}$
 $= \text{cl cone } \{y_i\}_{i \in I}$

Theorem: The conical hull of a finite set of vectors is a polyhedral cone.

Corollary: The convex hull of a finite set of points is a polytope.

Proof: Suppose $C = \text{cone } S$, where S is finite.

Then $C^0 = \{ y \in V^* : y(x) \leq 0 \quad \forall x \in S \}$

Since S is finite, C° is a polyhedral cone in V^* .
Thus $C^\circ = \text{cone } T$ for some finite set $T \subset V^*$.

$$C^{\circ\circ} = (\text{cone } T)^\circ$$

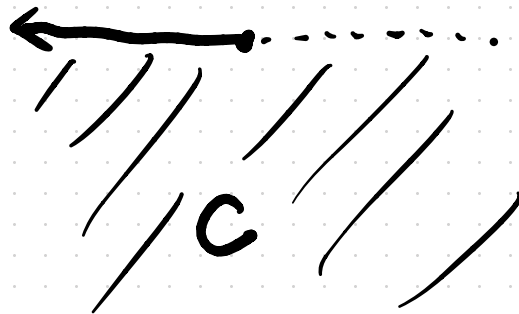
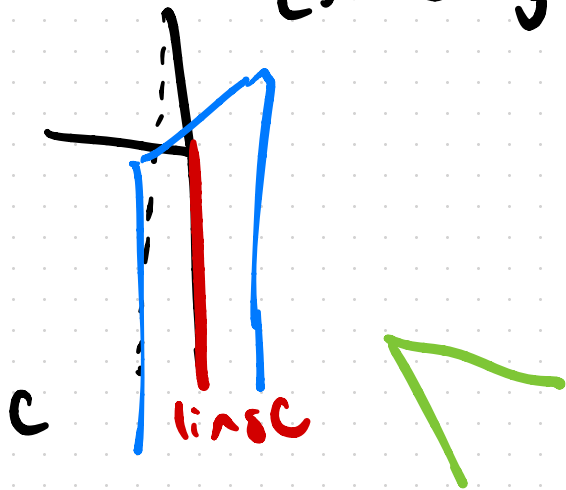
$$\underbrace{cl\ C}_{C} = \underbrace{\{x \in V : y(x) \leq 0 \ \forall y \in T\}}_{\text{polyhedral cone}}$$

since the conical hull
of a finite set is closed.

We say a cone C has **balanced support** if there is a linear functional $y \in V^*$ such that

$$y(x) \leq 0 \quad \forall x \in C \quad (y \in C^\circ)$$

$$\{x \in C : y(x) = 0\} = \text{lin } C$$



does not
have balanced
support

Theorem: If C is a closed convex cone, then it has balanced support.

Proof: Let $y \in \text{rint } C^\circ$. We claim this y satisfies the condition of balanced support.

relative interior, always nonempty

Since $y \in C^\circ$, we have $y(x) \leq 0$ for all $x \in C$.

This implies $y(x) = 0$ for $x \in \text{lin } C$, because

if $y(x) < 0$ for some $x \in \text{int} C$, then
 $-x \in C$ and $y(-x) > 0$, a contradiction.

Therefore

$$\text{int} C \subset \{x \in C : y(x) = 0\}$$

Need to show

$$\{x \in C : y(x) = 0\} \subset \text{int} C$$

Suppose $x \in C$ such that $y(x) = 0$.

Since C is closed, $C = C^{\circ\circ}$. Therefore
 $y'(x) \leq 0$ for all $y' \in C^\circ$.

Also $y(x) = 0$ for some $y \in \text{rint } C^\circ$.

There is a relative neighborhood of y in $\text{span } C^\circ$
which is contained in C . Since x evaluates to 0
on y and ≤ 0 on this neighborhood, it evaluates
to 0 on the neighborhood.



Therefore x evaluates to 0
everywhere on $\text{span } C^\circ$.

Exercise: For any closed core C ,

$\text{span } C$ and $\text{lin } C^\circ$ are orthogonal complements

$\text{lin } C$ and $\text{span } C^\circ$ are orthogonal complements.

Therefore $X \in (\text{span } C^\circ)^\perp = \text{lin } C$

Cor: If C is a closed convex cone with $\text{lin} C = \{0\}$,
then there exist $v \in V$ such that $v, -v \notin C$. $\dim C \geq 2$

